Majorana Modes in the Machine

Block 1: The Physics Bridge

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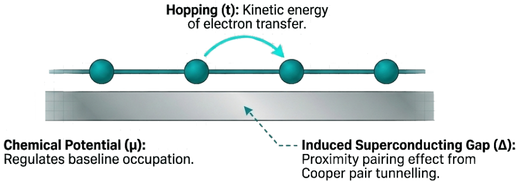
Outline

Starting point: the lattice Kitaev Hamiltonian

$$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + \Delta \sum_j (c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.}) \tag{1}$$

Physical ingredients

- Spinless (or spin-polarized) fermions
- Nearest-neighbor hopping t
- p-wave pairing $\Delta \in \mathbb{R}$
- Chemical potential μ tunes filling



Questions we will answer

- When does topology emerge?
- How do edge zero modes appear?

Majorana Representation & The "Sweet Spot"

Ordinary fermions can be split into two Hermitian **Majorana operators**:

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{2}$$

By rewriting H_K in terms of Majoranas (see Appendix for details), we find a special limit:

Sweet spot: $\mu = 0$, $t = \Delta$

The intra-site coupling vanishes. Only inter-site coupling $i\gamma_{B,j}\gamma_{A,j+1}$ survives. In a chain with open boundaries, this leaves **one unpaired Majorana zero-mode at each end**.

Why is this interesting?

These edge Majorana modes are **robust** (protected by topology) and candidates for **fault-tolerant quantum computation**.

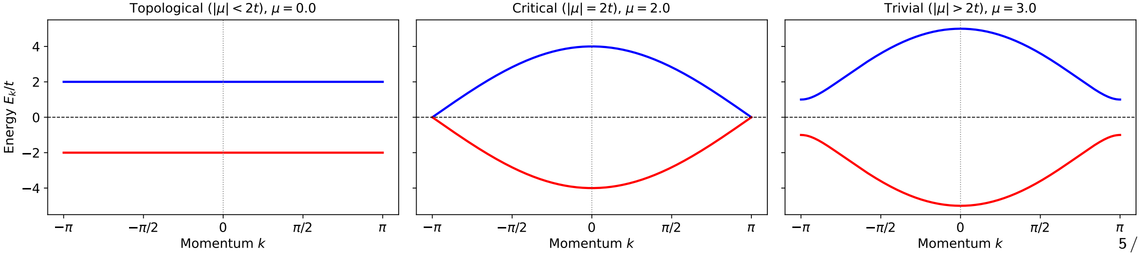
Bulk dispersion: topological / critical / trivial

Applying a Fourier transform and using the BdG formalism (see Appendix), the energy spectrum is given by:

$$E_k = \pm\sqrt{n_z(k)^2 + n_y(k)^2} = \pm\sqrt{(-\mu - 2t\cos k)^2 + (-2\Delta\sin k)^2} \tag{3}$$

The topological phase transition occurs when the **bulk gap closes** ($E_k = 0$). This happens at $k = \{0, \pi\}$ when $\mu = \pm 2t$.

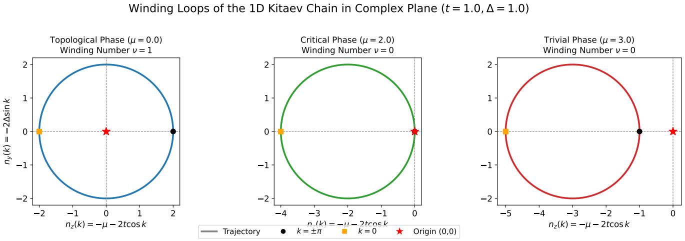
Bulk Dispersion of the 1D Kitaev Chain ($t = 1.0, \Delta = 1.0$)



Winding number ν

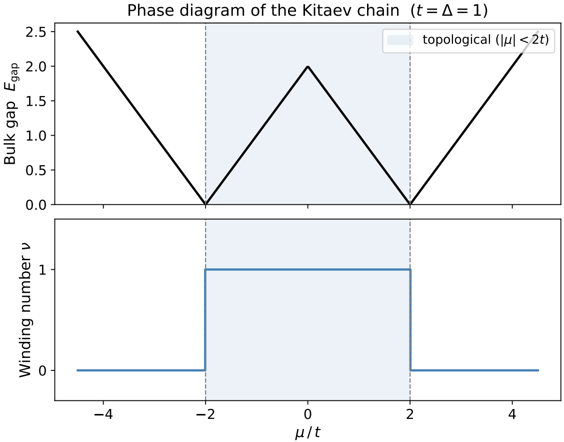
As k sweeps the BZ, the vector $\mathbf{n}(k) = (n_z, n_y)$ traces a closed loop.

$$\theta_k = \arg(n_z + i n_y), \qquad \nu = \frac{1}{2\pi}\int_{\mathrm{BZ}} \partial_k \theta_k \, dk \tag{4}$$



- $\nu = 0$: loop does *not* enclose origin $\rightarrow$ trivial
- $|\nu| = 1$: loop winds once around origin $\rightarrow$ topological

Phase diagram



Bulk gap vanishes at $\mu = \pm 2t$; winding number jumps across the transition.

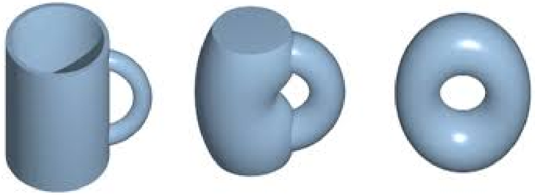
Topology: Continuity and Invariance

- **Topological Equivalence**: Two shapes are equivalent if they can be transformed into each other via “continuous deformation” (stretching, bending).
- **Invariants**: Properties that remain unchanged during deformation (e.g., number of holes / Genus).
- **Physical Link**: Tuning Hamiltonian parameters (t, μ) is essentially a “continuous deformation” of the wave function.

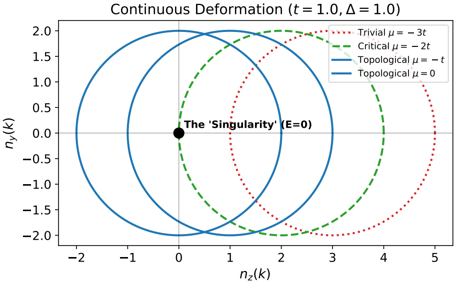
Core Intuition

You cannot turn a Sphere (0 holes) into a Donut (1 hole) without “tearing” the surface (Gap Closing).

Topological invariant: The Winding Number



with Genus 1



winding number $\nu = 1$

What is Protection?

As long as the circle doesn't cross the origin (Gap remains open), parameter fluctuations only "deform" the circle but cannot change the **integer** winding number.

Block 1 — key takeaways

1. The Kitaev Hamiltonian hosts **two phases** separated by gap closings at $\mu = \pm 2t$.
2. Majorana representation reveals the **sweet-spot** picture directly ($\mu = 0,\ t = \Delta$).
3. The BdG d-vector winding number $\nu \in \{0, 1\}$ is the **topological invariant**.

Next (Block 2): Finite size & edge modes $\rightarrow$ Jordan–Wigner transform $\rightarrow$ qubit encoding $\rightarrow$ quantum circuit for ground-state preparation.

Appendix

Appendix A: Full Majorana Representation

Recall the Majorana operators:

$$\gamma_{A,j} = \frac{c_j + c_j^\dagger}{\sqrt{2}}, \qquad \gamma_{B,j} = \frac{i(c_j^\dagger - c_j)}{\sqrt{2}} \tag{5}$$

where $\{\gamma_{\alpha,i}, \gamma_{\beta,j}\} = 2\delta_{\alpha\beta}\delta_{ij}$.

Rewriting the full Kitaev Hamiltonian in the Majorana basis yields:

$$H_K = -\mu \sum_j \Big(i\gamma_{A,j}\gamma_{B,j} + \tfrac{1}{2} \Big) + i \sum_j \Big[(\Delta + t)\, \gamma_{B,j}\gamma_{A,j+1} + (\Delta - t)\, \gamma_{A,j}\gamma_{B,j+1} \Big] \tag{6}$$

Note how setting $\mu = 0$ and $t = \Delta$ simplifies this to just the $i\gamma_{B,j}\gamma_{A,j+1}$ term.

Appendix B: Momentum Space Mapping

The tight-binding Hamiltonian in real space:

$$H = \sum_j \left[-t(c_j^\dagger c_{j+1} + \text{h.c.}) - \mu \left(c_j^\dagger c_j - \frac{1}{2} \right) + \Delta(c_j c_{j+1} + c_{j+1}^\dagger c_j^\dagger) \right] \tag{7}$$

Applying the Fourier transform $c_j = \frac{1}{\sqrt{N}} \sum_k e^{ikj} c_k$, the kinetic and chemical potential terms diagonalize trivially. The pairing term requires antisymmetrization due to fermionic statistics ($c_k c_{-k} = -c_{-k} c_k$):

$$\Delta \sum_k e^{ik} c_k c_{-k} = \Delta \sum_{k>0} (e^{ik} - e^{-ik}) c_k c_{-k} = 2i\Delta \sum_{k>0} \sin(k) c_k c_{-k} \tag{8}$$

This isolates the odd-parity p-wave superconducting nature of the model in momentum space.

Appendix C: BdG Formalism and Pseudospin

Use the Nambu spinor $\Psi_k = \begin{pmatrix} c_k \\ c^\dagger_{-k} \end{pmatrix}$ to cast H into a bilinear form:

$$H = \sum_{k>0} \Psi^\dagger_k h(k) \Psi_k, \quad h(k) = \begin{pmatrix} \xi_k & 2i\Delta \sin(k) \\ -2i\Delta \sin(k) & -\xi_k \end{pmatrix} \tag{9}$$

where the normal-state dispersion is $\xi_k = -\mu - 2t\cos(k)$.
Expanding $h(k)$ in Pauli matrices:

$$h(k) = n_y(k)\sigma^y + n_z(k)\sigma^z \implies \begin{cases} n_z(k) = -\mu - 2t\cos(k) \\ n_y(k) = -2\Delta \sin(k) \end{cases} \tag{10}$$

Diagonalizing this yields the bulk dispersion $E_k = \pm\sqrt{n_z(k)^2 + n_y(k)^2}$.